

Question	Marking Guidance	Mark	Comments
03.1	Lowercase a in both boxes	1	
03.2	Tick in box next to 'Crossing over';	1	
03.3	32.73 / 32.7 / 32 / 33;; Award 1 max for either 409 (409.2) for difference in volume (but incorrect number of mitochondria); OR Answer of 262 (261.9) (using diameter, rather than radius);	2	
03.4	1. Egg (created) has nucleus / DNA / genes of (affected) woman / mother; 2. It has mostly / many / lots of normal mitochondria (of unaffected woman) OR There are few faulty mitochondria;	2	1. Accept ref. to zygote / embryo / child for egg 1. Accept genetic information 1. Ignore references to alleles 1. Reject if nucleus from wrong egg / woman 2. Reject ref. to production of healthy mitochondria as result of treatment
03.5	1. Not enough / little ATP produced; 2. ATP provides energy for (enzyme) reactions OR ATP phosphorylates substrates / enzymes, so making them (more) reactive;	2 max	One reason asked for, so list rule applies 1. Ignore ref. to no ATP produced 2. Accept (leads to) lower activation energy for reaction 2. Reject if mention energy produced

